

3(a)	<u>aorta</u> / dorsal aorta ; <i>correct spelling needed</i>	1
3(b)(i)	any one from: context is using gaps for movement between, hepatocyte / tissue fluid, and, blood / sinusoid I ref. to red blood cells I shorter diffusion, distance / pathway higher rate of / easier for / AW, exchange / movement / AW, of substances ; A named substances / nutrients substances do not have to cross, a cell layer / into and out of endothelial cells ; larger molecules / plasma proteins, can pass through ; Kupffer cells can pass through (into tissue fluid) ; AW A ref. to other phagocytic cells R Kupffer cells can diffuse through R if stated as entering hepatocytes	1

Question	Answer	Marks
3(b)(ii)	<i>max 3 if described as phagocytosis of a pathogen rather than a red blood cell</i> <i>allow rbc for red blood cell</i> any four from: 1 endocytosis / phagocytosis ; 2 described ; e.g. red blood cell, engulfed / enveloped pseudopodia surround red blood cell <i>max three from</i> 3 forms (phagocytic) vacuole / (phagocytic) vesicle / phagosome ; 4 lysosome fuses (with phagosome) ; AW merges / joins to / combines with I attaches / binds to 5 (lysosome contains) hydrolytic / digestive, enzymes ; A hydrolases / named examples 6 digestion to produce haem and globin ; A amino acids for globin 7 AVP ; e.g. suggestion of, detection / recognition, of red blood cell	4

Question	Answer	Marks
3(c)(i)	any three from: I Golgi body / lysosome / vesicle <i>mp1 and mp2 complete term needed either in the text or labelled</i> 1 <u>smooth endoplasmic reticulum</u> labelled on Fig. 3.2 and detail ; e.g. for, lipid / triglyceride / cholesterol, synthesis tubular appearance / no ribosomes 2 <u>rough endoplasmic reticulum</u> labelled on Fig. 3.2 and detail ; e.g. for protein, synthesis / AW A for enzyme synthesis flattened (sac) appearance ribosomes A ribosome(s) labelled <u>and</u> for (plasma) protein synthesis 3 mitochondrion labelled on Fig. 3.2 <u>and</u> detail ; e.g. many mitochondria for (aerobic) respiration produce ATP / provide energy, (for hepatocyte reactions) R generate / make, energy double membrane / presence of cristae <i>allow if one mitochondrion is correctly labelled and one of the 'circular' mitochondria has a different label e.g lysosome</i> 4 presence of glycogen granule(s) as, energy / glucose, store ; A releases / provides, glucose 5 presence of lipid droplet(s), as energy store ; A provides fatty acids A provides components for cholesterol synthesis <i>penalise once (from mp1, 2, or 3) if diagram not labelled but structures named correctly and correct detail given</i>	

Question	Answer	Marks
3(c)(ii)	any one from: I ref. to staining or size mitochondrion, bound by / has, double membrane / two membranes ; I peroxisome has (only) one membrane peroxisome no cristae / mitochondrion has cristae ; R cisternae peroxisome spherical shape <u>and</u> mitochondria, different shapes / AW ; I if only peroxisome shape stated A circular <i>for spherical</i> A description e.g. 'sausage' shape	1
3(c)(iii)	any two from: <i>mitochondrion (ora for peroxisomes)</i> has, DNA / genes (coding for enzymes) ; R linear DNA A genetic, material / information, qualified with, transcription / ref. to mRNA / coding for enzymes (70S) ribosomes (site of enzyme synthesis) ; R 80S ribosomes has the enzymes for, transcription / translation, (to synthesise enzymes) ; can produce ATP (for enzyme synthesis) ; R produce energy I has ATP AVP ; ; <i>other suggestions</i> e.g. has (chaperone) proteins involved in, folding / formation of tertiary structure ATP cannot enter peroxisome to provide energy source peroxisome has enzymes that cause degradation of, one / or more, substance(s) involved in protein synthesis	2

Question	Answer	Marks
3(c)(iv)	any two from: (peroxisome) membrane bound, so rest of cell protected (from hydrogen peroxide) ; A hydrogen peroxide can have toxic effects on (rest of) cell / AW (peroxisome) acts as a compartment / specific area / contained, for, more efficient breakdown of hydrogen peroxide / control of reactions ; high concentration of enzyme in one location ; A more enzyme-substrate complexes can form provides, optimum / AW, conditions, for (other peroxisome) reactions ; production of oxygen as by-product, so useful for, peroxisome reactions requiring oxygen / mitochondrial function / aerobic respiration ; AVP ; e.g. isolates (peroxisomes) enzymes from rest of cell	2

Question	Answer	Marks
----------	--------	-------